The Effect of Forest Age on Machine Learning Evapotranspiration Generalisation

Author
Archie Benn

Supervisor
Professor Martin De Kauwe

Abstract (200 words)
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Key words: Evapotranspiration, Forest Age, Machine Learning, FLUXNET.
Word count: 4000 (Literature Review: 3000; Project Plan: 1000)

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science by advanced study in Bioinformatics in the Faculty of Health and Life Sciences.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
May 8, 2026

Contents

Introduction

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, and is a crucial component of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere (Katul et al. 2012). Approximately 60% of precipitation on the land surface is returned to the atmosphere through ET (Oki and Kanae 2006), and the associated latent heat flux, λET (where λ is the latent heat of vaporisation), accounts for 50% of the solar radiation absorbed by the land surface (Trenberth et al. 2009). ET can be generally described through:

$$ET = T + E_{Soil} + E_{Canopy} \tag{1}$$

where T refers to transpiration through plant stomata, E_{soil} to evaporation from the soil surface, and E_{Canopy} to evaporation from plant canopies (via rainfall interception) (Lawrence et al. 2007). The contribution of these variables to overall ET is not equal, with Lian et al. (2018) estimating a global T/ET ratio of 0.62±0.06, indicating plant transpiration asLikewise, the leading component of ET worldwide. The partitioning of ET into E ($E_{soil} + E_{Canopy}$) and T within ecosystems is associated with "water use efficiency" and has use in irrigation management, where T is linked with plant productivity, while E does not directly contribute to production (Kool et al. 2014). Accurate partitioning estimates are also important in aiding carbon cycle projections given the coupled effect of transpiration and land-atmosphere carbon dioxide exchanges through stomata.

Forest Complexity

The magnitude of ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed (Yang et al. 2023). Of these, Lian et al. (2018) identified LAI as the primary driver of T/ET spatial patterns, with Kool et al. (2014) suggesting that in systems with full canopy cover (and therefore high LAI) ET is often assumed to be "similar enough to T to permit correlation of ET with biomass". However, this assumption may break down in densely vegetated systems with multi-layer canopies such as forests, despite high LAI. In these systems, the 2-dimensional remote measure of LAI will not capture the vertical canopy complexity which drives rainfall interception and subsequent E_{Canopy}, implying that canopy cover indices may struggle to explain T/ET ratios in more complex vertical environments.

This is reflected by S. Zhou et al. (2016) who report higher coefficients of determination between enhanced vegetation index (EVI) and T/ET in corn and soybean sites (0.84 and 0.82) with simple canopies, compared to evergreen needle leaf forests (0.37), a result possibly due to increased E_{Canopy} driving this ratio down. Hardiman et al. (2013) also demonstrate that while LAI may be stable in stands older than 50 years, canopy complexity continues to increase with age through >160 years of stand development and had a greater impact on annual net primary productivity (NPP) and overall resource use efficiency than LAI across all stands. Furthermore, LAI has shown to only explain around 20% of mean annual T/ET (X. Li et al. 2019), suggesting simple vegetation indices which don't take into account 3-dimensional complexity may be insufficient for predicting T/ET in structurally complex systems.

Oishi et al. (2020), in a chronosequence study on broadleaf forests in the US, report that interception of incoming rainfall strongly increases with age from ~2% (12 year old stand) to 20% (200 year old stand) despite only small differences in LAI, an effect in part due to branch architecture which varies with age. This study also reported that annual canopy transpiration (T) was over twice as high in the 35 year old stand compared to the 12 and 200 year old stands, with increasing canopy interception in the oldest stand largely offsetting the reduction in T such that total ET was similar between the two older stands, but remained lower in the youngest stand (Figure 1). This demonstrates how age-driven effects in canopy architecture introduce complexity and can affect overall ET, as well as the relative contributions of T and E_{Canopy} which may vary significantly over time within stands.

Alongside canopy architecture, stomatal conductance (g_s) declines as trees grow taller due to gravitational reductions in leaf water potential and decreasing hydraulic conductance with increasing path length from soil to leaf (Ryan and Yoder 1997). For example, Schäfer et al. (2000) demonstrated that the mean stomatal conductance of tree crowns decreased by approximately 60% over a 30m increase in tree height, and Hubbard et al. (1999) established a 44% drop in hydraulic conductance in old trees compared to young trees in a mixed age stand. As g_s directly influences T (Monteith 1965), these age related changes introduce further complexity into ET dynamics and suggest that age could be a key source of variability in ET in forest ecosystems.

Forest Age

Forest age distributions are increasingly being affected by climate change-driven disturbances - such as fire, drought, insect attack, and pathogens - which are expected to continue in the coming decades (Solomon et al. 2009; Seidl et al. 2017). Some of these disturbances are able to completely change forest demography by causing widespread mortality (Dale et al. 2001), effectively resetting the age of the forest landscape. Land use changes also directly affect forest age distributions, for example the recovery of deforested areas in southern China following the implementation of forestation policies around the year 2000 has resulted in a complex pattern of forest age structure in recent years, and an increase in average tree cover from 21%

in 1987 to 38% in 2016, leading to large areas now being dominated by young forests (Tong et al. 2020; Leng et al. 2024), with similar patterns of young forests due to recent disturbance found worldwide (Besnard et al. 2021).

As forests are exhibiting greater age heterogeneity, and with climate change increasing the severity and frequency of potential forest disturbances worldwide, stand age may represent an increasingly important variable for ET estimation in forests, capturing age specific effects such as canopy complexity and alterations to g_s, and linking these to ET beyond what simple vegetation indices such as LAI are capable of. Yet whether these age-driven changes in forests are a large enough source of ET variability to affect the estimations made by current ET models which do not take age into account remains largely unexplored.

Evapotranspiration Prediction

Why is ET Predicted?

Estimating ET accurately is important in understanding hydrological processes, management of irrigation systems for effective water use in agriculture, prediction and mitigation of droughts, determining the adaptive mechanisms of vegetation, and in assessing the overarching patterns of global climatic shifts (Amani and Shafizadeh-Moghadam 2023; Y. Xue et al. 2025). These assessments can, in turn, steer regulations and policies in climate science and water resource management, representing a critical link between the quantification of terrestrial water fluxes and land systems governance (Haiyang Shi et al. 2024).

The direct measurement of ET is the most accurate and uses lysimeters, eddy covariance (EC), or sap flow, but these are costly to use and only measure at the plot scale, resulting in spatially constrained measurements which makes them unsuitable for describing ET at larger ecosystem scales (Amani and Shafizadeh-Moghadam 2023; Liyew et al. 2025). The uneven distribution and low sampling rate of EC data in ecoregions of Africa, Oceania (excluding Australia), and South America may be attributed to the costs associated with these methods, despite efforts to mitigate this (Hill et al. 2017; Markwitz and Siebicke 2019). As a result of these issues, various methods to estimate ET indirectly in areas without the capacity for direct measurement have been derived, including: (1) traditional process-based models which rely on pre-defined equations such as the Penman-Monteith (PM) and Hargreaves-Samani (HS) equations (Monteith 1965; Hargreaves and Samani 1985); (2) remote sensing (RS) - based models, for example the SEBS model (Z. Su 2002); and more recently, (3) machine learning (ML) methods such as artificial neural networks and random forest models trained on historical data (T. Xu et al. 2018). The development of these methods has permitted global ET estimates to be produced using more easily acquired variables such as air temperature and relative humidity (S. Pan et al. 2020), yet significant uncertainties remain, particularly in underrepresented regions and ecosystems with sparse EC coverage, as well as in complex or heterogeneous landscapes (T. Xu et al. 2018).

ET Estimation Methods

There are many different methods for the estimation of ET as no single model performs reliably across all conditions and not all are similarly accurate or reliable for different regions (Derardja et al. 2024). Thus the data requirement inputs can be significantly varied across models (Zhao et al. 2013). Process-based models define the relationship between input variables and ET explicitly for estimations which are developed through hydrological theory, while ML approaches do not explicitly define relationships, instead relying on the underlying architecture of models to optimise against the provided training data to derive the relationship between inputs and ET. RS data cannot provide direct estimations of ET, but instead estimate retrievable surface parameters associated with ET such as vegetation indices like LAI, surface temperatures, and soil moisture (Amani and Shafizadeh-Moghadam 2023). For example, Lilin Zhang et al. (2021) demonstrated improved ET estimations in water-limited agroecosystems by using satellite retrievals as alternatives to the soil evaporation component of the process-based Priestley-Taylor Jet Propulsion Laboratory (PT-JPL) model.

While process-based methods such as the PM equation are able to achieve high accuracy across climatic zones, they suffer from a requirement of comprehensive input meteorological data and their rigid structure limits adaptability, for example their calibration is typically time and space dependent which restricts generalisation (Amani and Shafizadeh-Moghadam 2023). The continuous and broad spatial monitoring of variables affecting ET has been facilitated through advances in RS technology, yet RS methods are limited by low per pixel resolution (e.g 30m^2 for Landsat and 500m^2 for MODIS) which affects fine scale spatial prediction accuracy, alongside issues with temporal coverage from cloud interference and long revisitation times (16 days for Landsat) (J. Xue and B. Su 2017).

ML methods have been used extensively for ET estimation in recent years and have demonstrated high spatiotemporal prediction accuracy and a capability to outperform process-based and RS-based models (S. Pan et al. 2020; Yan et al. 2023). Their increased precision of ET estimations in data rich regions may be due to their ability to leverage both linear and non-linear relationships for mapping between input data and ET estimates (Y. Xue et al. 2025). In data limited contexts, coupled ML models which use process-based model outputs as engineered input features have shown improved performance compared to standalone methods. For example, Yan et al. (2023) deployed a HS-ML coupled model which incorporated the HS model output as an input feature for a long short-term memory neural network (LSTM) and demonstrated better performance compared to the standalone HS and ML methods in predicting daily potential ET (ET_o), while Y. Liu et al. (2021), in a reverse approach, used ML to predict g_s before applying this value to the PM equation and demonstrated greater performance over the PM equation alone. Traditional (non DL) ML approaches, such as Random Forest (RF) and Support Vector Machine (SVM), are commonly used and also show good ET predictive performance, with T. Xu et al. (2018) comparing the capabilities of five ML models against EC data as ground truth, finding the RF model to slightly outperform the others in uncertainty, although all models predicted ET to high accuracy (R^2 0.87 – 0.89). LTSMs are a specialised form of recurrent neural network (RNN), itself a deep learning (DL) architecture falling under the broader term artificial neural network (ANN), which are highly capable in extracting long term dependencies within time series data and handling noisy datasets representing a potential method for broad application and precise estimations in ET (Yan et al. 2023).

ML methods, while promising in reducing uncertainties and improving ET estimation accuracy, can struggle with extrapolation beyond training data and lack the interpretability of process-based models, often being termed 'black boxes' due to their inherent complexity and lack of decision making explainability (Hassija et al. 2024). The quality of ML ET estimations is highly dependent on the provided training data, model hyperparameter tuning, and choice of predictors (feature selection) - all of which relies on user knowledge and pre-existing data for the specific ecosystem in question (Amani and Shafizadeh-Moghadam 2023). Therefore careful consideration must be taken during ML training and evaluation to form a relevant and accurate ET estimation model for the specific use case.

Feature Selection

The selection of features in ET estimation models varies from a minimal data input (**X**) to a broad choice of variables (**Y**). The range of selected features is often constrained by the availability of meteorological data in the target region, but also varies during attempts to form models which can be applied to data limited regions. Table 2 presents selected ET prediction models within studies along with the chosen input variables for each method.

Machine Learning and Applicability of Incorporating Age ~800 Words

(What is the case for using ML and forest age here?)

Why use Machine Learning?

Why Incorporate Age? + how on larger scale in future? - RS methods for tree demography?

Effect on Generalisation

Quick section closing off the review and what the project will aim to investigate

Project Aims and Objectives (500 words)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

Project Plan (500 words)

Data Acquisition

Site Selection

Analysis

Feature Engineering

ML Model Setup and Training

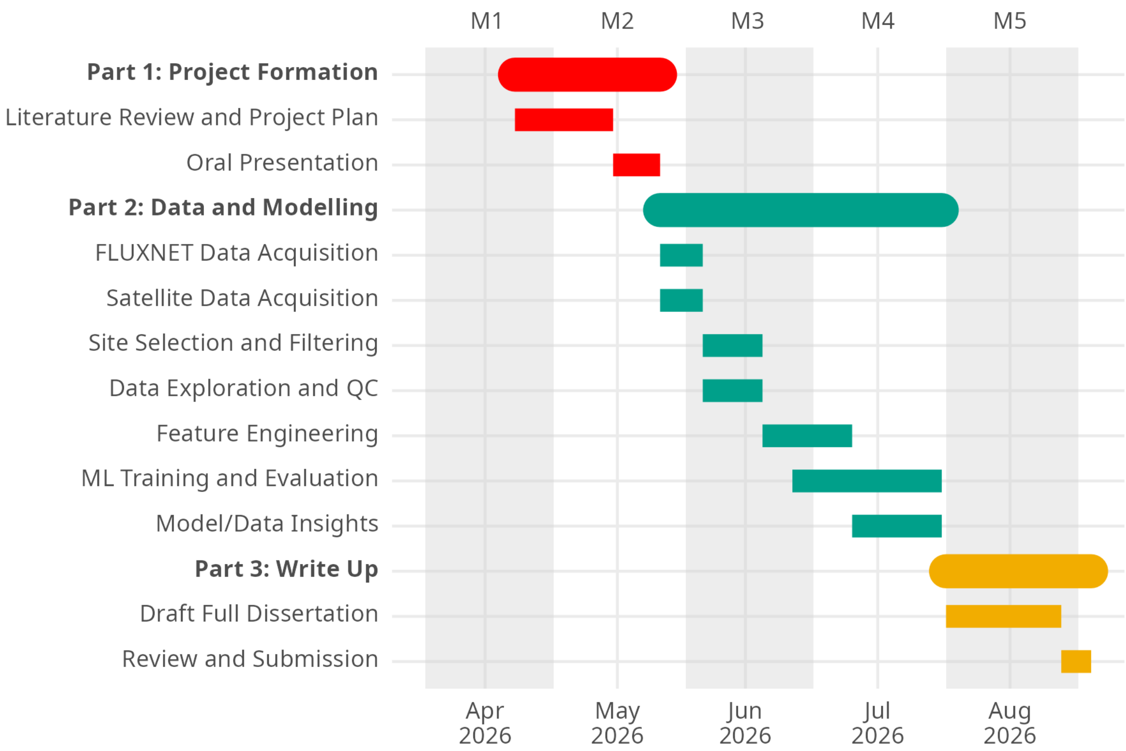
ML Model Evaluation

ML Model Predictions

Tools

Interpretation

Gantt Chart


M1
M2
M3
M4
M5
Part 1: Project Formation
Literature Review and Project Plan
Oral Presentation
Part 2: Data and Modelling
FLUXNET Data Acquisition
Satellite Data Acquisition
Site Selection and Filtering
Data Exploration and QC
Feature Engineering
ML Training and Evaluation
Model/Data Insights
Part 3: Write Up
Draft Full Dissertation
Review and Submission
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026

Risk Register (~ 250 words)

Table 1: Risk register

Nature of the risk	**Likelihood and potential impact on the project**	**Mitigation strategy**
ahhh	oh	lipsum
yeh	risk	register

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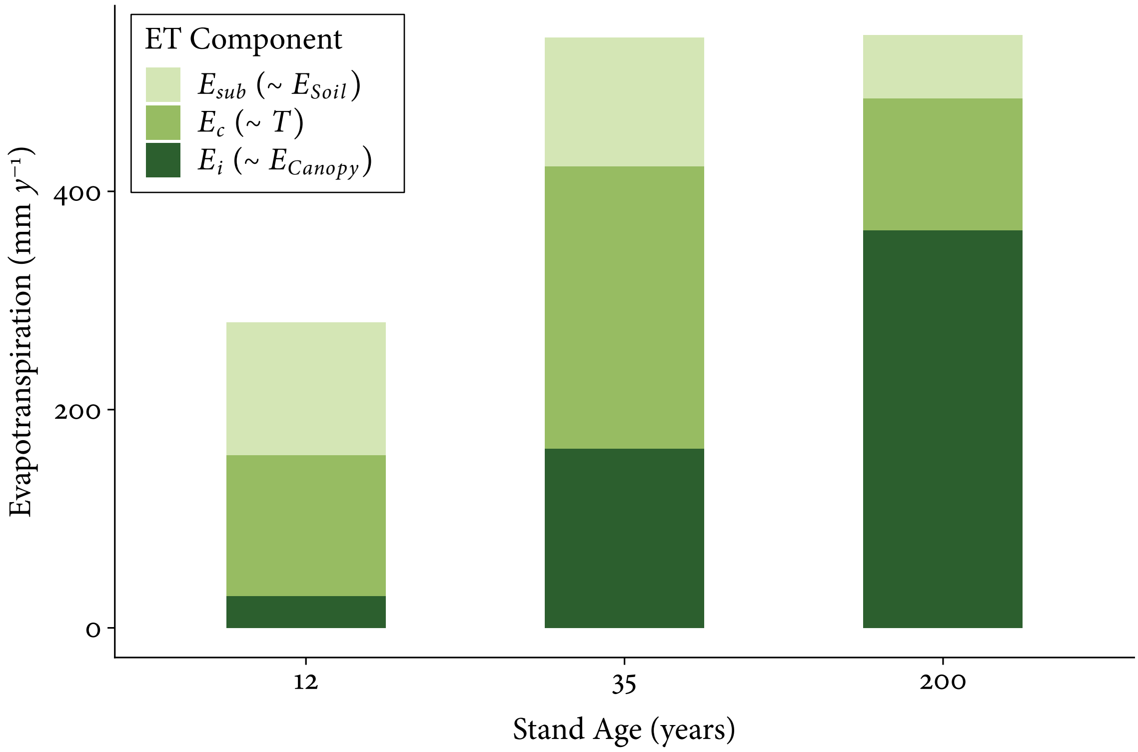

Figure 1: Components of annual evapotranspiration by stand age, redrawn from Oishi et al. (2020). Sub canopy evapotranspiration (E_{sub}), canopy transpiration (E_c), and canopy interception (E_i) correspond approximately to E_{Soil}, T, and E_{Canopy} in Equation 1, although E_{sub} additionally captures under canopy transpiration.

Table 2: Advantages and limitations of ET models in the literature.

Model	Input Features	Ecosystem	Appraisal
Random Forest (T. Xu et al. 2018)	Outperformed other models (ANN, cubist, SVM, DBN) based on R^2, MAPE, RMSE, and relative uncertainties from three-cornered hat. RF has an ability to handle complex non-linear relationships between flux and climate and environmental predictors.	Model performs better over areas of dense vegetation compared to barren land. More spatiotemporal continuous land surface variables at fine resolution should be added to potentially enhance predictability.	yeh
HS-LSTM coupled model (Yan et al. 2023)	Shown to outperform the standalone ML and classical ET_0 models (RMSE reduced ~20%). Can be used to estimate future ET_0 with only air temperature forecasts which can be easily obtained.	Slight underestimates of extreme ET_0 values. Each station required a differently-tuned model, so further studies are required to evaluate potential for a generally valid model and assess whether transfer learning may be applied across different regions.	yeh